

GLMs for Applied Research

Led by Eliana Ibrahimi, University of Tirana

Session 4: Poisson Regression for Count Data

Topics we will cover in this session

- Recap: Why linear regression breaks down for counts?
- The Poisson distribution & the log-link model
- Fitting and interpreting Poisson regression
- Detecting and correcting overdispersion

What is count data?

- Non-negative integers/event occurrences $(0, 1, 2, 3, \dots)$.
 - hospital admissions/month,
 - ED visits/week,
 - medication errors/unit/month,
 - patient falls/ward/month.
- Bounded at zero, discrete, often right-skewed.

The Poisson regression model

Modeling counts

$$\log(\mu_i) = \beta_0 + \beta_1 X_{1i} + \dots + \beta_k X_{ki}$$

where $\mu_i = E(Y_i | X)$ is the expected count for observation i .

The log link guarantees $\mu_i = \exp(\dots) > 0$ for any combination of predictors — predicted counts can never be negative.

Modeling rates: the offset

$$\log(\mu_i) = \log(t_i) + \beta_0 + \beta_1 X_{1i} + \dots$$

t_i is exposure — time, population, or area — with its coefficient fixed at 1.

Use an offset whenever observations differ in the time, area, or population at risk of the event.

Core assumptions

- Independent observations
- Correctly specified linear predictor & log link
- $\text{Var}(Y) = E(Y)$

Poisson Regression: Interpretation

Rate Ratio (Incidence Rate Ratio)

$$IRR_j = e^{\beta_j}$$

- A one-unit increase in X_j **multiplies** the expected count (or rate) by e^{β_j} holding other predictors constant.
- MLE estimation, deviance-based fit, AIC for model selection, likelihood ratio tests for nested models, the same `glm()` interface in R.

β	$IRR = e^{\beta}$	Interpretation
0	1.0	No effect
0.41	1.5	50% more events
-0.22	0.8	20% fewer events

Example

Term	Log scale (β)	$\exp(\beta)$	Interpretation
Intercept	-0.20	0.82	Expected count when all predictors = 0 / reference
Age (per year)	0.045	1.046	4.6% increase in expected count per additional year
Treatment (yes vs. no)	-0.51	0.60	40% reduction in expected count for treated group

$\exp(\beta)$ = the multiplicative change in the expected rate for a one-unit increase in X , holding other predictors constant.

Rule of thumb: $\exp(\beta) > 1$ means the rate increases; $\exp(\beta) < 1$ means it decreases; $\exp(\beta) = 1$ means no association.

```
model <- glm(  
  outcome ~ predictor1 + predictor2 + ...,  
  offset = log(exposure),  
  family = poisson(link = "log"),  
  data = your_data  
)  
  
summary(model)      # coefficient table  
exp(coef(model))    # rate ratios  
exp(confint(model)) # 95% CI, rate-ratio scale
```

Syntax notes

outcome ~ predictors

Any count variable on the left; any mix of numeric or categorical predictors on the right.

offset = log(exposure)

Include only when observations differ in time, area, or population at risk. Omit entirely otherwise.

family = poisson(link="log")

Standard Poisson. Swap to quasipoisson(link="log") if the dispersion check flags overdispersion.

Overdispersion

The Poisson model assumes $\text{Var}(Y) = E(Y)$. Real-world count data is frequently “overdispersed”: $\text{Var}(Y) > E(Y)$.

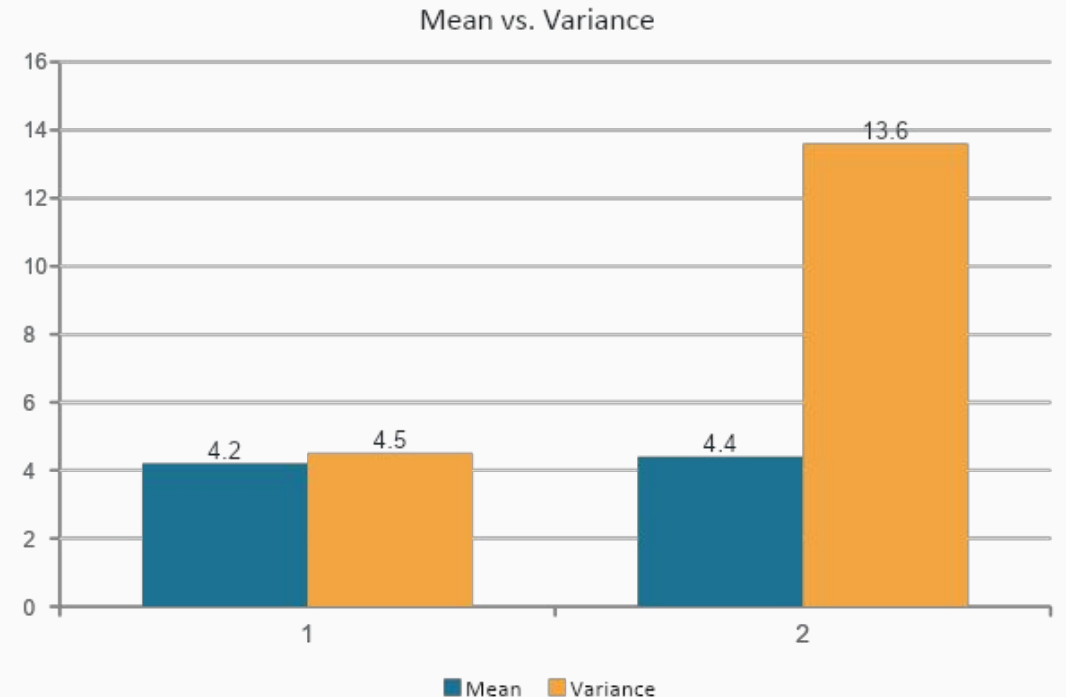
Dispersion statistic

$$\hat{\phi} = \text{Residual deviance} / \text{df}$$

$\hat{\phi} \gg 1 \rightarrow \text{overdispersion present}$

Common causes

- Unobserved heterogeneity between subjects
- Clustering or correlation within groups
- Excess zeros beyond what Poisson predicts



Overdispersion: Solutions

- Quasi-Poisson: inflates standard errors by $\sqrt{\phi}$ — same coefficients, correct inference
- Negative Binomial regression: adds an individual-level random effect; models $\text{Var}(Y) = \mu + \mu^2/\theta$ (preferred approach)
- Zero-inflated models: when zeros are more frequent than Poisson predicts

Model building strategies:

- Likelihood Ratio Test (LRT): test whether adding a predictor significantly improves fit: $-2(\ell_{\text{reduced}} - \ell_{\text{full}}) \sim \chi^2(\text{df})$
- AIC / BIC: lower is better; BIC penalizes complexity more heavily — useful when parsimony matters
- Stepwise selection: automated but use with caution — can overfit and doesn't account for multiple testing

Model diagnostics

Issue	How to detect	Fix
Influential observations	Cook's distance, leverage	Investigate outliers
Overdispersion (Poisson)	Deviance/df $\gg 1$	Quasi-Poisson or Negative Binomial
Multicollinearity	VIF > 10	Drop/combine predictors
Non-linear effects	Residual plots, splines	Add polynomial or spline terms

Residuals

- **Pearson residuals:** $(\text{Observed} - \text{Fitted}) / \sqrt{\text{Var}(\text{Fitted})}$
- **Deviance residuals:** contribution of each observation to the total deviance
- Plot both against fitted values and predictors to check for patterns.

Recap & What's next

- Log link keeps predictions non-negative
- Coefficients exponentiate to rate ratios
- Add offsets to model rates
- Check dispersion before trusting SEs
- Use quasi-Poisson/NB when overdispersed

Next session: Hands-on - Modeling a count outcome